

Pursuit-Evasion Games II

Tomáš Kroupa

AI Center
Department of Computer Science
Faculty of Electrical Engineering
Czech Technical University in Prague

2023

Two-Player Zero-Sum Games

Summary

What we have learned

- Matrix games capture the strategic interaction between two players
- Their solutions are found by LP
- Efficient algorithms
- Single stage games

Where we will go

- Dynamic games
- Iterative interaction between players in the changing environment
- Scalability issues

Hierarchy of Dynamic Games

- Partially observable stochastic games (states not fully observable)
- Stochastic games
  1. Repeated normal-form games (more players and 1 state)
  2. Markov Decision processes (1 player and many states)

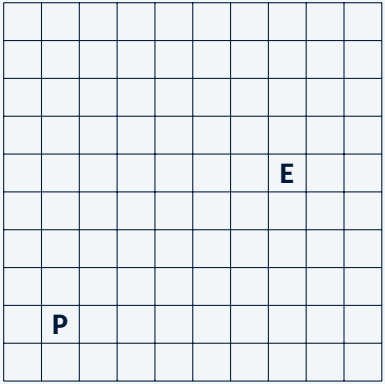
There are other game forms for modelling dynamics:

- Extensive-form games with imperfect information
- Stackelberg games

Stochastic Games

Repeating Zero-Sum Games

👍 Or making MDPs competitive?



- Pursuer **P** tries to capture evader **E**
- Stochastic policy describes the mixed strategy of each player in every state
- We are seeking a common generalization of
  - repeated TPZS games and
  - Markov decision processes (MDPs)

Show two examples here

1. Players are the planner and the adversary
2. Finite action sets M and N for the planner and the adversary, respectively
3. Finite set S of states
4. Transition function

$$T : S \times M \times N \to \Delta_S$$

where $T(s, i, j)$ is a probability distribution on S

5. Reward function

$$R : S \times M \times N \to \mathbb{R}$$

where $R(s, i, j)$ is a reward to the planner

- The general theory of SG is fraught with subtleties
- For example, the reward function may also depend on the state entered from the current state $R\colon S \times M \times N \times R \to \mathbb{R}$

How to Play a Stochastic Game?

A sequence of zero-sum stage games

An initial state is $s_0 \in S$ at stage 0.

At each stage $\tau = 0, 1, 2, \ldots$:

1. The players choose $(i_\tau, j_\tau) \in M \times N$ observing the complete history of past states/actions
2. The planner receives $R(s_\tau, i_\tau, j_\tau)$ from the adversary
3. A new state $s_{\tau+1} \in S$ is determined randomly according to $T(s_\tau, i_\tau, j_\tau)$

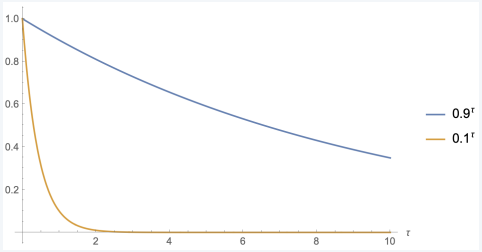
How to aggregate the rewards over an infinite history?

Discounting

The importance of future rewards

- A discount factor is a number $\gamma \in (0, 1)$
- The discounted reward of planner over an infinite history $(s_0, i_0, j_0, \dots)$ is

$$\sum_{\tau=0}^{\infty} \gamma^{\tau} R(s_\tau, i_\tau, j_\tau) = R(s_0, i_0, j_0) + \gamma R(s_1, i_1, j_1) + \gamma^2 R(s_2, i_2, j_2) + \cdots$$



Note that γ is a part of the learning objective, not a tunable parameter of some algorithm!

Since there are only finitely many values of rewards r_τ, we get

$$\sum_{\tau=0}^{\infty} \gamma^\tau r_\tau \le r \sum_{\tau=0}^{\infty} \gamma^\tau = r \frac{1}{1-\gamma}.$$

Two interpretations of γ:

1. discount factor (small for myopic agents, large for farsighted)
2. $1-\gamma$ is the probability that game terminates, so $\gamma^{T-1}(1-\gamma)$ is the probability that game terminates after T rounds

Strategies

For stochastic games

At stage τ the players observe an entire history

$$h_\tau := (s_0, i_0, j_0, \ldots, s_\tau, i_\tau, j_\tau).$$

A strategy of the planner is

1. behavioral if it is a mapping $h_\tau \mapsto p \in \Delta_M$ from histories to mixed actions
2. Markov if it is a behavioral strategy depending only on the current state s_τ
3. stationary if it is a Markov strategy not depending on time τ

Stationary Strategies

A stationary strategy for the planner/adversary is a mapping

$$\pi : S \to \Delta_M \quad \text{and} \quad \sigma : S \to \Delta_N,$$

respectively.

Special cases

- For $|S| = 1$ we get the concept of mixed strategy in a repeated two-player zero-sum game
- If $|N| = 1$, then we get an MDP with the concept of stochastic policy

How to evaluate stationary strategies?

Value/Quality Function

☞ $(s_\tau), (i_\tau), (j_\tau)$ are stochastic processes with respect to T, π, σ

- The value function is the expected reward starting from state $s_0 = s$ and using strategies π, σ:

$$\mathcal{V}_{\pi,\sigma}(s) = \mathbb{E}_{T,\pi,\sigma}\left(\sum_{\tau=0}^{\infty} \gamma^\tau R(s_\tau, i_\tau, j_\tau)\right).$$

- The Q-function (or action-value function) is the expected reward starting from state s, taking actions i, j, and then following strategies π, σ:

$$\mathcal{Q}_{\pi,\sigma}(s,i,j) = R(s,i,j) + \mathbb{E}_{T,\pi,\sigma}\left(\sum_{\tau=1}^{\infty} \gamma^\tau R(s_\tau, i_\tau, j_\tau)\right).$$

Equilibrium of a Stochastic Game

Maximin/minimax

- A stationary strategy π^* of the planner is a maxmin strategy if

$$\min_{\sigma} \mathcal{V}_{\pi^*,\sigma}(s) = \max_{\pi} \min_{\sigma} \mathcal{V}_{\pi,\sigma}(s) \qquad \text{for every } s \in S$$

- A stationary strategy σ^* of the adversary is an minmax strategy if

$$\max_{\pi} \mathcal{V}_{\pi,\sigma^*}(s) = \min_{\sigma} \max_{\pi} \mathcal{V}_{\pi,\sigma}(s) \qquad \text{for every } s \in S$$

- A pair (π^*, σ^*) is an equilibrium

Theorem (Shapley, 1953)

Every two-player zero-sum stochastic game has an equilibrium.

Quality/Value Functions for Equilibria

For equilibrium strategies (π^*, σ^*) define

- the equilibrium value function

$$\mathcal{V}_*(s) := \mathcal{V}_{\pi^*,\sigma^*}(s) = \max_{\pi} \min_{\sigma} \mathcal{V}_{\pi,\sigma}(s) = \min_{\sigma} \max_{\pi} \mathcal{V}_{\pi,\sigma}(s)$$

- the equilibrium Q-function

$$\mathcal{Q}_*(s,i,j) := \mathcal{Q}_{\pi^*,\sigma^*}(s,i,j)$$

Our goal is to obtain one-step ahead formula replacing the infinite sum.

Expectation Equations

For any pair of strategies π and σ:

$$\mathcal{V}_{\pi,\sigma}(s) = \sum_{i \in M} \sum_{j \in N} \pi_s(i) \cdot \sigma_s(j) \cdot \mathcal{Q}_{\pi,\sigma}(s,i,j)$$

$$\mathcal{Q}_{\pi,\sigma}(s,i,j) = R(s,i,j) + \gamma \sum_{s' \in S} T(s,i,j)(s') \cdot \mathcal{V}_{\pi,\sigma}(s')$$

We will use the second equation for equilibrium strategies iteratively:

$$\mathcal{Q}_{*}(s,i,j) = R(s,i,j) + \gamma \sum_{s' \in S} T(s,i,j)(s') \cdot \mathcal{V}_{*}(s')$$

Value Iteration for Stochastic Games

Baseline algorithm

1. Initialize $\mathcal{V}$ arbitrarily
2. For each state $s \in S$ and all action pairs $(i,j) \in M \times N$ do

$$\mathcal{Q}(s,i,j) \leftarrow R(s,i,j) + \gamma \sum_{s' \in S} T(s,i,j)(s') \cdot \mathcal{V}(s')$$

3. For each state $s \in S$ do

$$\mathcal{V}(s) \leftarrow \max_{\pi_s \in \Delta_M} \min_{j \in N} \sum_{i \in M} \pi_s(i) \cdot \mathcal{Q}(s,i,j)$$

4. Repeat 2-3 until $\mathcal{V}$ converged

- This is a direct generalization of value iteration for MDPs
- It is model-based: requires access to T and R
- At step 2 each $\mathcal{Q}(s, ., .)$ is a "discounted stage game", which is just a matrix game
- At step 3 the value $\mathcal{V}(s)$ of this game is computed

Minimax Q-learning

Set a learning rate $\alpha \in (0, 1)$.

1. Initialize $\mathcal{Q}, \mathcal{V}$ arbitrarily
2. For each state $s \in S$ and all action pairs $(i, j) \in M \times N$ do

$$\mathcal{Q}(s, i, j) \leftarrow (1 - \alpha)\mathcal{Q}(s, i, j) + \alpha(R(s, i, j) + \gamma \mathcal{V}(s))$$

3. For each state $s \in S$ do

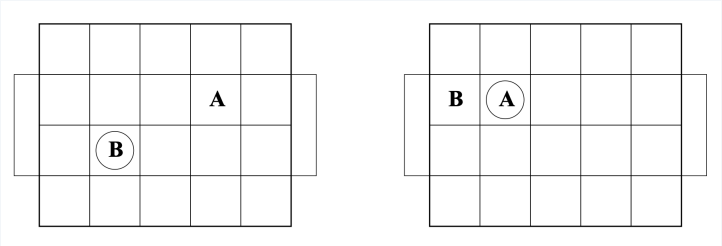
$$\mathcal{V}(s) \leftarrow \max_{\pi_s \in \Delta_M} \min_{j \in N} \sum_{i \in M} \pi_s(i) \cdot \mathcal{Q}(s, i, j)$$

4. Repeat 2-3 until $\mathcal{V}$ converged

Examples

Soccer (Littman, 1994) with $\gamma = 0.9$

- 780 states (players' position and ball possession)
- 5 actions (N, S, E, W, stand)
- Random change of ball possession and the action selection



Examples

Goofspiel (Flood, 1930s)

- Each player and the deck have one suit of cards numbered $1, \ldots, n$
- A card from the deck is bid on secretly
- The player with the highest card gets the corresponding reward
- Each player discards the card bid
- Repeat for n rounds

n	$\lvert S \rvert$	$\lvert S \times A \rvert$	SIZEOF(π or Q)	V(det)	V(random)
4	692	15150	$\sim$ 59KB	-2	-2.5
8	3×10^6	1×10^7	$\sim$ 47MB	-20	-10.5
13	1×10^{11}	7×10^{11}	$\sim$ 2.5TB	-65	-28

References

- Albrecht, S. V., Christianos, F., and Schäfer, L. "Multi-agent reinforcement learning: Foundations and modern approaches." Massachusetts Institute of Technology: Cambridge, MA, USA, 2023.
- Bowling, Michael, and Manuela Veloso. "Scalable learning in stochastic games." AAAI Workshop on Game Theoretic and Decision Theoretic Agents. 2002.
- Littman, Michael L. 1994. Markov Games as a Framework for Multi-Agent Reinforcement Learning. *Machine Learning Proceedings*, 1994. https://doi.org/10.1016/b978-1-55860-335-6.50027-1.